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An active learning representative subset selection method using net analyte signal

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ABSTRACT

To guarantee accurate predictions, representative samples are needed when building a calibration model for spectroscopic measurements. However, in general, it is not known whether a sample is representative prior to measuring its concentration, which is both time-consuming and expensive. In this paper, a method to determine whether a sample should be selected into a calibration set is presented. The selection is based on the difference of Euclidean norm of net analyte signal (NAS) vector between the candidate and existing samples. First, the concentrations and spectra of a group of samples are used to compute the projection matrix, NAS vector, and scalar values. Next, the NAS vectors of candidate samples are computed by multiplying projection matrix with spectra of samples. Scalar value of NAS is obtained by norm computation. The distance between the candidate set and the selected set is computed, and samples with the largest distance are added to selected set sequentially. Last, the concentration of the analyte is measured such that the sample can be used as a calibration sample. Using a validation test, it is shown that the presented method is more efficient than random selection. As a result, the amount of time and money spent on reference measurements is greatly reduced.

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1. Introduction

Spectroscopy offers many advantages over conventional analytical methods, such as no sample preparation requirements, the capacity for simultaneous measurement of multiple components and a non-destructive nature. However, spectroscopy is a secondary reference method in which the measured spectra are related to the reference values obtained using a primary reference method through multivariate calibration approaches such as partial least squares regression (PLSR). To obtain adequate calibration models, good calibration data and efficient estimation procedures are required. The requirements for a calibration dataset can be summarized into two factors [1]: a sufficiently large sample size and representative samples that cover the concentration range of the future samples to be analyzed. Existing sample selection methods can be categorized into three classes, i.e., y-reference-excluded, y-reference-included, and active learning.

The main advantage of y-reference-exclude methods is the lack of a requirement for reference measurements (that is, the methods are unsupervised), which saves much time and money. Random selection (RS) [2], Kennard-Stone (KS) (also called the computer-aided design of experiments, CADEX) algorithm [3–4], D-optimality [5], successive projections algorithm (SPA) [6], aim to maximize the Euclidean distances between the instrumental response vectors of the selected samples. One disadvantage of unsupervised selection methods is that the selected samples are based on the aggregated spectral effects of all analytes in a spectrum and not specific analyte-related information. The selected sample may not be a typical sample for a specific analyte. There is a risk that differences between minor analytes may be overwhelmed by major constituents, particularly in complex solutions.

Sample set partitioning based on joint x–y distances (SPXY) [7] and reference value (YR)-based sample selection method [8] are typical y-reference-included calibration methods. It has been proven [8] that y-reference-included selection methods perform better than y-reference-excluded selection methods because the former ones are based on analyte-related information. So in order to select representative samples for different analytes, analyte-related (supervised) information should be included in the sample selection procedure. However, the costs of y-reference measurements are higher than those of X, i.e., spectrum measurements. A calibration set of n samples implies not only n spectra but also n reference measurements of a given response variable.

The optimal solution to this dilemma is to find dependent (y)-related information using fewer reference measurements. As a result, active learning sample selection methods should be used. The idea of active learning strategies is based on adding samples that are distant from the current training samples in the feature space [9–12]. The main challenge associated with the distance lies in determining how to select a feature space that is representative of different concentrations.

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The net analyte signal (NAS) is defined as the part of the measured signal that the model relates to the analyte concentration [13]. A net analyte vector is uniquely related to the concentration of the analyte of interest. The NAS approach is capable of translating spectral information to a concentration-related metric. Because the NAS vector can provide important information about the main spectral features of the analyte of interest, it is used to represent the dependent information about the samples in this study. In the proposed method, concentration-related knowledge (NAS vector) is first obtained using labelled samples (samples with analyte concentration measured) from a regression model. Then, analyte-related information in candidate samples (unlabelled samples, without concentration) is obtained by active learning. Finally, more informative samples, determined by the difference in distance of the NAS vector's norm, are selected for inclusion in the calibration set.

2. Sample Selection Method

2.1. Notation

Throughout the present work, matrices are noted in capital bold (as in $\mathbf{X}$), column vectors in small bold (as in $\mathbf{x}$), vector and matrix elements in nonbold and indices in subscript characters (as in X_{ij} or x_i), and index limits in nonbold, italicized capital characters as in I or J. Row vectors are shown as transposed vectors (as $\mathbf{x}^{\mathbf{T}}$). The dimensions of some relevant matrices are as follows: calibration data matrix $\mathbf{X}$, $I \times J$ (I = number of calibration samples and J = number of wavelength), vector of calibration concentrations or reference values for analyte k ($\mathbf{c_k}$), $I \times 1$, and vector of analytical responses for an unknown sample $\mathbf{x}$, $J \times 1$.

2.2. Definition of NAS

Algebraically, the NAS can be defined as the part of the signal orthogonal to the space spanned by the interferences (all components except for the analyte of interest), as shown in Fig. 1, which was originally proposed by Lorber [13]. The NAS projection splits the data matrix $\mathbf{X}$ into two matrices: one containing the net spectral information pertaining to the analyte k, $\mathbf{X_k}$, and the other containing the spectral information pertaining to all of the other variability sources, $\mathbf{X_{-k}}$, including the contribution of the interferents ($\mathbf{X = X_k + X_{-k}}$). The NAS vector for a given sample is calculated by projecting the spectrum $\mathbf{x}$ onto the space defined by the interferents (matrix $\mathbf{X_{-k}}$), with the NAS being the orthogonal resultant ($\mathbf{X_k}$).

2.3. Computation of the Net Analyte Signal for the Inverse Calibration Case

In an inverse calibration [14] situation, the NAS calculation is performed by solving a rank annihilation problem. The main steps and formula are listed below. Detailed descriptions can be found in references [15,16].

(1) Perform singular value decomposition (SVD) on response matrix $\mathbf{X}$

$$[\mathbf{U\,S\,V}] = \mathrm{svd}(\mathbf{X}) \tag{1}$$

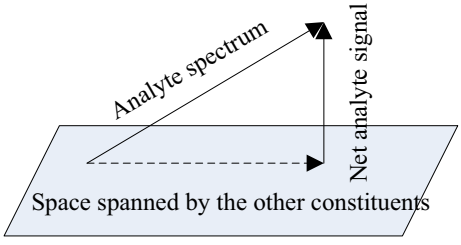


Fig. 1. Net analyte signal, as the part of the analyte spectrum that is not spanned by the spectra of the other components.

(2) Rebuild $\mathbf{X}$ using only A significant components, yielding the matrix $\hat{\mathbf{X}}$

$$\hat{\mathbf{X}} = \mathbf{U_A S_A V_A^T} \tag{2}$$

$\mathbf{U_A}$, $\mathbf{S_A}$, and $\mathbf{V_A}$ are submatrices with A significant components of the corresponding vector computed by SVD to $\mathbf{X}$. A was determined based on the amount of spectral variance explained. The significant components that accounted for less than 1% of the total spectral variance were ignored. In our experiment, component number is 5.

(3) $\hat{\mathbf{c}}_\mathbf{k}$ are the concentration vectors $\mathbf{c_k}$ projected onto the A-dimensional space

$$\hat{\mathbf{c}}_\mathbf{k} = \hat{\mathbf{X}}\hat{\mathbf{X}}^{+}\mathbf{c_k} \tag{3}$$

the superscript "+" indicates the Moore-Penrose pseudo-inverse for the generalized inverse and can be computed from the SVD matrix

$$\mathbf{X}^{+} = \mathbf{V_A S_A^{-1} U_A^T} \tag{4}$$

(4) The scalar α is computable as

$$\alpha = \mathbf{1}/\hat{\mathbf{x}}^{\mathbf{T}}\hat{\mathbf{X}}^{+}\hat{\mathbf{c}}_\mathbf{k} \tag{5}$$

where $\hat{\mathbf{x}}$ is row mean of $\hat{\mathbf{X}}$.

(5) The rank annihilation step in the A-dimensional space is given by

$$\hat{\mathbf{X}}_{-\mathbf{k}} = \hat{\mathbf{X}} - \alpha\hat{\mathbf{c}}_\mathbf{k}\hat{\mathbf{x}}^{\mathbf{T}} \tag{6}$$

(6) The projection matrix for kth analyte is

$$\mathbf{H} = \mathbf{I} - \hat{\mathbf{X}}_{-\mathbf{k}}^{\mathbf{T}}\left(\hat{\mathbf{X}}_{-\mathbf{k}}^{\mathbf{T}}\right)^{+} \tag{7}$$

where $\mathbf{I}$ is the J-dimensional identity matrix and $\mathbf{H}$ is a J-dimensional square.

(7) The net analyte signal of the kth analyte for an unknown sample is then calculated by

$$\hat{\mathbf{x}}_{\mathbf{k,un}}^{*} = \mathbf{H}\mathbf{x}_\mathbf{un} \tag{8}$$

(8) The scalar value (Euclidean norm) representation is

$$\widehat{\mathrm{NAS}}_{\mathrm{k,un}} = \left\|\hat{\mathbf{x}}_{\mathbf{k,un}}^{*}\right\| \tag{9}$$

2.4. Active Learning Sample Selection Procedure

The NAS active learning selection procedure is a deterministic sequential approach that attempts to select samples separately distributed in the response space. The NAS procedure for selecting a calibration subset of n samples ($\mathbf{X_{tr}} = \{X_{trj}\}_{j=1}^{n}$) from a given set of N samples ($\mathbf{X} = \{X_i\}_{i=1}^{N}$, note that $n < N$) consists of the following steps. The selection steps are also illustrated in Fig. 2.

(1) A part of the sample set is randomly selected, and the analyte concentrations in each sample are measured, i.e., labelled. This set is denoted $\mathbf{X_L}$, where L is the sample number.

(2) The NAS value NAS_L of $\mathbf{X_L}$ is computed using the algorithm described in Section 2.2 by combining spectra $\mathbf{X_L}$ and analyte concentrations $\mathbf{c_L}$.

(3) The NAS values $\mathrm{NAS_U}$ of the candidate (unlabelled, having U samples) samples $\mathbf{X_U}$ ($L + U = N$) are computed by simple multiplication of the current spectrum vector $\mathbf{x_U}$ with the projection matrix $\mathbf{H}$. Eqs. (8–9) should be used.

(4) The samples are chosen sequentially. Supposing that k samples have already been selected ($L \le k \le N$), the $(k + 1)th$ sample in the

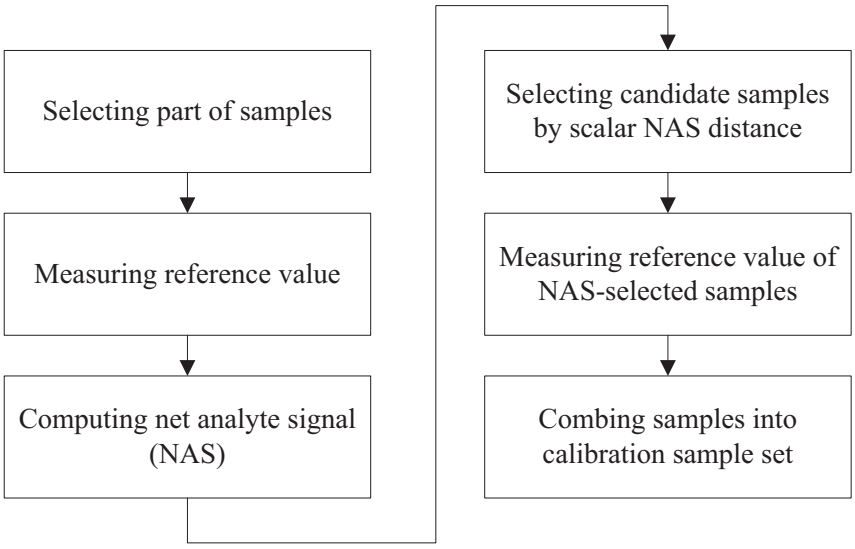


Fig. 2. Active learning sample selection method process.

training set is chosen using the criterion

$$\max_{1<r<N-k}\left(\min(d_{1r}, d_{2r}, \ldots, d_{kr})\right)$$

where $d_{jr} = |\|\hat{x}_r^{net}\| - \|\hat{x}_j^{net}\||$, $j = 1,\ldots, k$, are the absolute values of the NAS differences between a candidate object r not yet included in the representative set and the k objects already included in the representative set. The computations are repeated until a predefined number n reached.

(5) The analyte concentrations in the selected samples are measured using the reference method so that the samples can be used as calibration samples. The concentration measurements are performed only on the specific analytes that are selected.

(6) The calibration model is updated with the enlarged sample set.

3. Experiments

3.1. Sample Collection and Reference Measurement

Two sets, one was soy sauce, another was corn gluten meal, each have 300 samples randomly collected from two factories. Initially, 50 prediction samples were randomly extracted from the full set to simulate the analysis of a batch of real unknown samples. The remaining 250 samples were divided into a labelled set containing 50 samples used for NAS computation and candidate sets containing 200 samples to be selected.

Two important ingredients in soy sauce were modelled individually, i.e., amino acid nitrogen (AN) and total acid (TA), to illustrate the effect of representative samples on various calibration models. Moisture and protein contents were modelled for corn gluten meal. A reference analysis of TA and AN was performed by titration using methanal and NaOH standard solution, respectively. The moisture content was measured indirectly through dry mass determination at 105 °C, according to GB/T 12087-2008. The protein content was measured by the Kjeldahl method described in GB 5009.5-2010 using an NYK6160 analyser (Shanghai Yihon Analysis Instruments company).

3.2. Spectral Measurements and Preprocessing

Spectral data for the soy sauce samples were collected by transflectance in the NIR region of 950–1650 nm in 2 nm increments using a diode-array analyser (DA 7200, Perten Instruments, Sweden). Spectral data for corn gluten meal were measured by reflectance in 5 nm increments using same type of spectrometer and wavelength region. Duplicates of each sample were scanned twice. The average spectrum of each sample was used in the subsequent data analysis.

Typical spectra of soy sauce and corn gluten meal are shown in Fig. 3 (a) and (b), respectively. Fig. 3 (b) shows distinct baseline shifts in the raw spectra of the corn gluten meal set. This phenomenon is attributed to scattering due to different powder particle sizes. The baseline shifts were eliminated by a standard normal variate (SNV) preprocessing method. No preprocessing was performed for the soy sauce spectra. PLS is used in modelling setup.

3.3. Evaluation Parameters

The statistical parameters and calibration model performance are presented in Tables 1 and 2. In the tables, label represents the 50 label samples used to compute the representative NAS vector and scalar. Total represents the total samples (250 samples), including not only the labelled set but also the candidate set. The NAS set (100 samples) includes the labelled set and the selected representative samples. Random means that a calibration set have 100 samples, with additional samples selected randomly. The prediction set is an independent sample set with 50 samples randomly collected. SD indicates standard deviation. RMSE (C for calibration set, CV for cross validation, P for prediction set) is the root mean square error computed by the following formula:

$$\mathrm{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(y_i - \hat{y}_i)^2} \tag{10}$$

where M is the sample number, y_i is the sample true value, and $\hat{y}_i$ is the sample predicted value. The RMSEs were selected as indicators of model performance because they are the most direct parameters indicating model performance. The RMSEP is regarded as the most representative parameter of model accuracy.

4. Results and Discussion

4.1. Validation of NAS Representation

In order to prove the consistency of the NAS values obtained from other sample set predictions and their own predictions, 50 samples were selected in the candidate set to compare the NAS values calculated

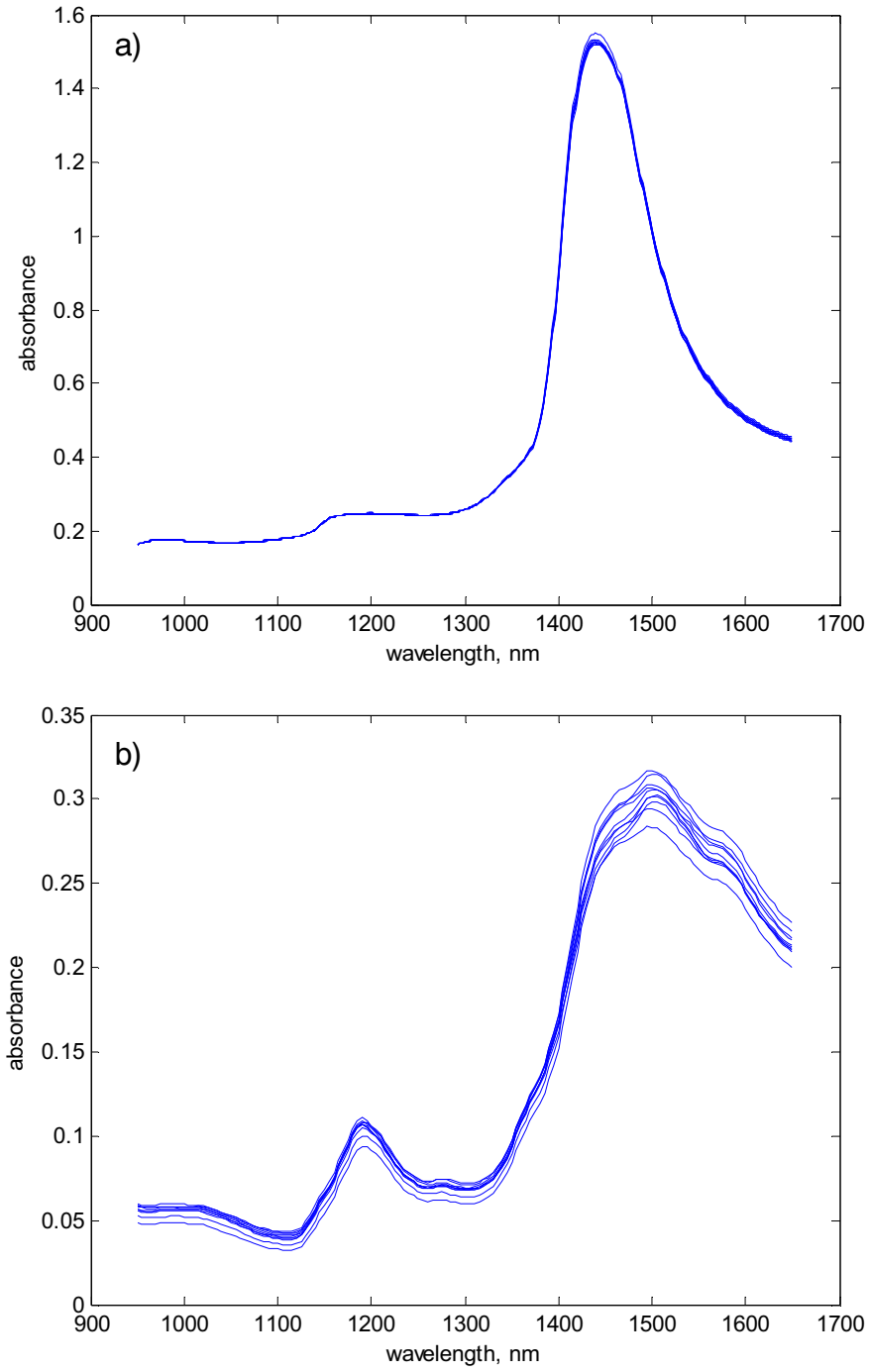


Fig. 3. Typical spectra of (a) soy sauce and (b) corn gluten meal.

Table 1
Statistical parameters and prediction results of the different sample sets for soy sauce (Unit: relative weight, %), amino acid nitrogen (AN) and total acid (TA) being analyte.

TA	Range	Mean	SD	RMSEC	RMSECV	RMSEP
Label	0.53–1.20	0.859	0.150	0.0665	0.0729	0.0748
Total	0.46–1.35	0.874	0.159	0.0777	0.0805	0.0599
NAS	0.46–1.21	0.859	0.168	0.0793	0.0834	0.0643
Random	0.53–1.23	0.925	0.154	0.0802	0.0873	0.0743
Prediction	0.64–1.32	0.941	0.163			
AN	Range	Mean	SD	RMSEC	RMSECV	RMSEP
Label	0.41–0.79	0.547	0.087	0.0221	0.0250	0.0207
Total	0.36–0.79	0.559	0.091	0.0149	0.0168	0.0201
NAS	0.41–0.79	0.553	0.088	0.0184	0.0202	0.0191
Random	0.41–0.79	0.558	0.092	0.0174	0.0191	0.0184
Prediction	0.47–0.75	0.604	0.060			

by itself (marked by circle "o") and the NAS values predicted by the calibration set (marked by cross "+") . The results are shown in Fig. 4. The mean difference of NAS ($\overline{\Delta NAS}$) between the two sets is 6.73e-004. The mean value of NAS ($\overline{NAS}$) is 0.3256. The mean relative difference of ($\overline{\Delta NAS}$)/($\overline{NAS}$) is 0.21%, which means that the samples have similar NAS values. This feature ensures that the representation of a sample by its scalar value is reasonable.

4.2. Prediction Result for Soy Sauce Set

For the two constituents of soy sauce, the model performance with NAS selection are presented in Table 1, and the RMSEP values are

Table 2
Statistical parameters and prediction results of the different sample sets for corn gluten meal (Unit: relative weight, %).

	Range	Mean	SD	RMSEC	RMSECV	RMSEP
Moisture						
Label	5.1–10.1	7.890	1.090	0.211	0.256	0.287
Total	5.1–10.9	7.823	1.014	0.292	0.306	0.262
NAS	5.1–10.9	7.950	1.172	0.257	0.304	0.271
Random	5.1–10.1	7.847	1.020	0.260	0.283	0.274
Prediction	6.4–10.2	7.998	1.010			
Protein						
Label	44.6–61.6	57.45	3.221	0.350	0.455	0.517
Total	44.4–62.8	57.62	3.633	0.414	0.463	0.471
NAS	44.4–62.8	56.797	4.132	0.435	0.515	0.486
Random	44.4–62.8	57.862	3.625	0.430	0.481	0.526
Prediction	53.5–61.9	58.522	2.293			

compared in the histogram shown in Fig. 5. The variation of RMSEP for AN is confined to a very limited range for all of the RMSE indices. Even when comparing the label and total sets, a smaller change can be observed. Based on the high accuracy of AN, it can be concluded that AN is an ingredient that is easily measured by spectroscopy, and the addition of samples to the calibration set has little effect on the model's predictive performance. In the following discussion, the relevant AN parameters are not included.

For TA, clear differences can be observed in Fig. 5 and Table 1. The RMSEP of TA is 0.0643 for the NAS set, whereas it is 0.0743 for the randomly selected set with the same number of samples as the NAS. An accuracy comparison demonstrated the superiority of the NAS method over random selection. Furthermore, NAS selection with fewer samples added (50 samples) approached the performance of the method involving a large sample (200 samples), with the RMSEP value of 0.0643 approaching the value of 0.0599. Compared with the accuracy of the total sample set, high efficiency in improving the accuracy was achieved using the NAS selection method.

The SD values of the NAS and random sets containing the same samples were compared. The SD values of TA (0.168 for NAS to 0.154 for random) were clearly different. The values for the NAS set were greater than those for the random set, indicating that the NAS set is more informative than the random set and is better for establishing a calibration model. The SD of the random set is similar to that of the labelled sample set, which indicates that the SD levels of both the randomly selected set and the labelled sample set are less than those of the NAS set.

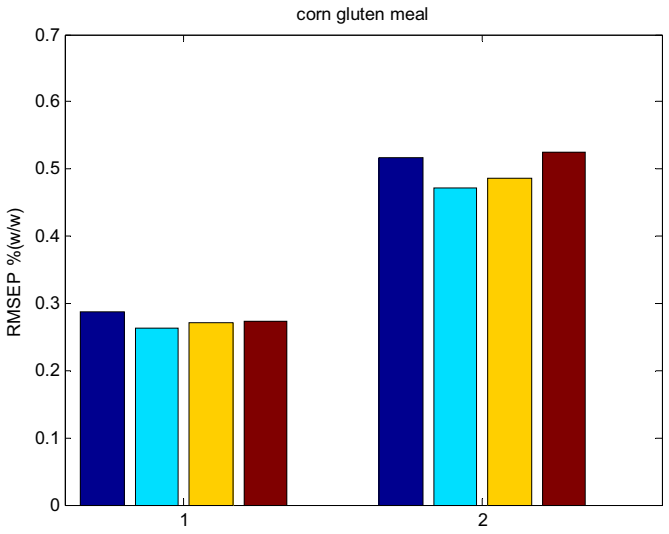


Fig. 6. RMSEP values of different sample sets. The two constituent groups from left to right are moisture and protein. The bars from left to right in every group are label, total, NAS, and random.

4.3. Prediction Results for Corn Gluten Meal

The prediction results for corn gluten meal using different sample sets are presented in Table 2 and Fig. 6. Based on the results, the RMSEP values pertaining to moisture content between the different sets are similar, although some differences can be observed. The main improvement is observed for protein content. The RMSEP value for protein content is 0.486 for the NAS set, whereas it is 0.526 for the random set. An accuracy comparison demonstrated the superiority of the NAS method over random selection.

It can be observed that the large calibration set (total set) has a small prediction error, 0.262 for moisture content and 0.471 for protein content; both values are significantly smaller than the error for the small calibration set (50 for labelled and 100 for NAS). The labelled set (50

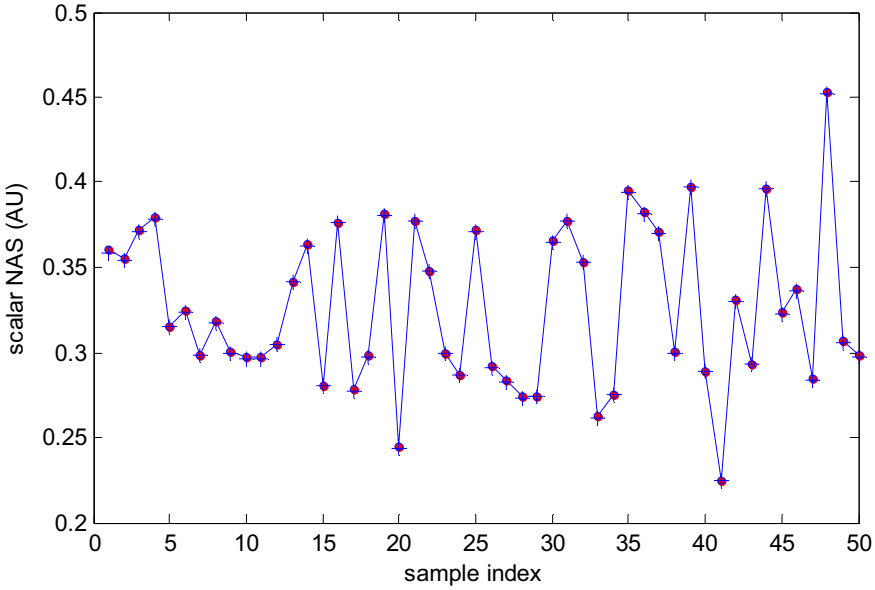


Fig. 4. NAS values of TA of the candidate set computed by its own set (marked by circle "°") and predicted by the labelled set (marked by cross "+") are compared. The mean relative difference $(\overline{\Delta NAS})/(\overline{NAS})$ is 0.21%. Similar NAS value can be observed, which indicates that the NAS of new samples can be deduced using the existing sample set.

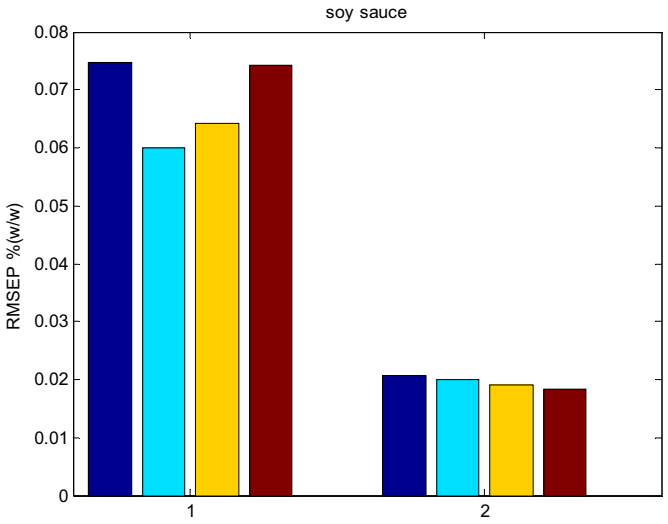


Fig. 5. RMSEP values of different sample sets. The two constituent groups from left to right are TA and AN. The bars from left to right in every group are label, total, NAS, and random.

samples) may have small RMSEC (0.350, protein as example) and RMSECV values (0.455), but have a large RMSEP value (0.517). The difference between the three parameters is rather large, which means the model prediction capability is poor. As a result, the number of samples significantly affects model performance, even though some representative samples have been selected. However, NAS selection with fewer samples added (50 samples) approached the performance of that with many samples added (200 samples), with the RMSEP value of 0.486 comparing well with the value of 0.471. Compared with the accuracy of the total sample set, high efficiency in improving the accuracy was achieved using the NAS selection method, similar to what was observed for the soy sauce set. With all 200 samples added, the RMSEP value was further reduced; however, more concentration measurements must be performed.

4.4. Significance Test

To determine whether the proposed method is significantly better than the random selection method, a significance test was performed. The significance test parameters are listed in Table 3. Selection is performed ten times ($n = 10$) for the sample set, and a *t*-test is performed. The level of significance (α) is set to 0.05. $\overline{d}$ is the mean value, and **s** is the standard deviation of the RMSEP difference for the selection. ***t*** is a test statistic computed by the equation $t = \frac{\overline{d}}{s/\sqrt{n}}$, $t_\alpha(n-1)$. It is the threshold value (1.8331) for $n = 10$, and $\alpha = 0.05$, and can be get from table. It can be observed that the elevations of prediction accuracy of both TA (in soy sauce) and protein (in corn gluten meal) using NAS selection are significant. The difference of accuracy is not significant for either AN (in soy sauce) or moisture content (in corn gluten meal).

Table 3
Significance test for RMSEP difference calibrated by NAS set and random set. Parameter meaning please refer to text.

analyte	$\overline{d}$	s	n	α	t	$t_\alpha(n-1)$	Obvious
TA	−0.0051	0.0074	10	0.05	−2.0951	1.8331	Yes
AN	−0.00072	0.0042			−0.4551		No
Moisture	0.0032	0.0245			0.3961		No
Protein	−0.0346	0.0529			−1.9624		Yes

5. Conclusions

The presented method for selecting samples using NAS computation effectively increases the representativeness of samples without y-value measurements compared with the random selection method. With the cost of measuring the concentration values for a smaller number of original samples, the distribution of scalar NAS values can be obtained. The subsequent candidate samples are evaluated by their NAS scalar norm values. A sample with the maximum NAS distance value from existing samples is selected into the calibration set to ensure its representativeness. Using this method, only the concentrations of the representative samples are submitted to measurement, which is both time- and cost-efficient. The sample selection method offers great advantages for online spectroscopic measurement, where spectra of many samples can be obtained conveniently; however, samples are very similar. It should be noted that although samples are selected by NAS, matrix effect remains the same to original sample set because no other factors are emphasized in the selection procedure.

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